Industry Dynamics

Competitive Industry Dynamics and Dynamic Oligopoly

October, 2026

Where We Are

- So far we have focused on demand and its implications for relatively **short-run** supply responses:
  - prices and quantities;
  - product configuration;
  - static mergers and vertical integration.
- In all of these the set of firms and their heterogeneity (costs, qualities) are taken as given.

- This lecture asks what determines the **long-run** market structure and firm heterogeneity:
  - *Static entry models* (Bresnahan–Reiss, Berry, Sutton): market structure and competition are determined jointly, but in a single shot.
  - *Dynamic models*: entry and exit, investment (physical capital, R&D, learning), dynamic pricing, and how the market structure evolves over time.

- IO economists treat these decisions as **strategic**. But firm turnover and heterogeneity matter well beyond IO: productivity growth, trade, misallocation, and business-cycle research all build on the same models.

Road-map

- **Part I — Facts.** What do the data say about entry, exit, growth, and the firm-size distribution? Gibrat's law and Zipf's law; Dunne, Roberts, and Samuelson (1989).
- **Part II — Selection with learning.** Jovanovic (1982): firms learn their own efficiency; the efficient grow and survive.
- **Part III — Competitive industry dynamics.** Hopenhayn (1992): a stationary equilibrium with simultaneous entry and exit; an application to Taiwan and Korea (Aw, Chung, and Roberts 2003).
- **Part IV — Dynamic oligopoly.** Markov perfect equilibrium; the Ericson and Pakes (1995) framework.
- **Part V — Computation.** Pakes and McGuire (1994) and its curse of dimensionality; the stochastic algorithm of Pakes and McGuire (2001); approximations: oblivious equilibrium and moment-based Markov equilibrium.
- **Part VI — Multiplicity.** Besanko, Doraszelski, Kryukov, and Satterthwaite (2010): learning-by-doing and organizational forgetting.

One trade-off runs through the lecture

A **continuum** of price-taking firms (Hopenhayn) makes the aggregate state deterministic and the model easy to solve, but rules out strategic interaction. A **finite** number of strategic firms (Ericson–Pakes) keeps the strategy, but the state space explodes.

Basic Facts of Firm Dynamics

Before formal models of *industry* dynamics, we document a few "consensus facts" about *firm* dynamics, the building blocks of an industry equilibrium.

- **Entrants**
  - High "infant mortality": the exit hazard is highest in the first years and declines steadily afterwards.
  - The *surviving* entrants grow faster than incumbents.
- **Exit**
  - Exit rates decline systematically with firm age and size.
  - Across mature industries, entry and exit rates are *positively* correlated $\Rightarrow$ common determinants (e.g., sunk costs) rather than net expansion or contraction.
- **Growth**
  - Gibrat's law holds roughly *at the top*: among large firms, growth rates are independent of size.
  - Small (young) firms grow faster than large (old) firms.
- **Heterogeneity**: large and persistent productivity differences across plants within narrowly defined industries.

Gibrat's Law and the Firm-Size Distribution

If firms grew mechanically according to Gibrat's law, what would the size distribution look like?

- Let size evolve as $s_{t+1} = \gamma_{t+1} s_t$, with growth factors γ_{t+1} i.i.d. with density $f(\gamma)$, independent of s_t.
- Let $G_t(x) \equiv \Pr(s_t > x)$ be the counter-cumulative distribution. Then

$$G_{t+1}(x) = \Pr(\gamma s_t > x) = \int_0^\infty \Pr\left(s_t > \frac{x}{\gamma}\right) f(\gamma)\, d\gamma = \int_0^\infty G_t\left(\frac{x}{\gamma}\right) f(\gamma)\, d\gamma.$$

- A steady-state distribution must satisfy $G(s) = \int_0^\infty G(s/\gamma) f(\gamma)\, d\gamma$.
- Conjecture a **power law** $G(s) = k\, s^{-\zeta}$. Substituting,

$$k\, s^{-\zeta} = \int_0^\infty k\, s^{-\zeta} \gamma^\zeta f(\gamma)\, d\gamma \quad \Longrightarrow \quad E[\gamma^\zeta] = 1.$$

- If the average firm does not grow, $E[\gamma] = 1$, then $\zeta = 1$ solves this equation: Zipf's law (Gabaix 2009). ($E[\gamma^\zeta]$ is convex in ζ and equals 1 at $\zeta = 0$, so $\zeta = 1$ is the only positive root.)

Caveat

Pure Gibrat growth has no steady state: $\log s_t$ is a random walk and its variance grows without bound. A power law emerges once we add a friction that stops firms from shrinking indefinitely, such as a lower reflecting barrier or **entry and exit**. That is where the economics comes in.

Firm-Size Distribution: Zipf's Law

- **Power law (Pareto) distribution**:

$$\Pr(S > s) = \left(\frac{s_0}{s}\right)^{\alpha}, \qquad s \geq s_0,\ \alpha > 0.$$

$\alpha = 1$ is called "Zipf's law". Zipf distributions approximate a wide range of natural and social phenomena (city sizes, word frequencies, incomes at the top).

- **Checking it with data**
  - Rank firms by size, $s_{(1)} \geq s_{(2)} \geq \cdots \geq s_{(n)}$.
  - The empirical survival function at $s_{(r)}$ is $\approx r/n$.
  - Under a power law, $\log(r/n) \approx \alpha \log s_0 - \alpha \log s_{(r)}$: a log-log plot of rank against size is a **straight line with slope** $-\alpha$.

- Axtell (2001) does this for the entire population of tax-paying U.S. firms (1997 Census): $\alpha \approx 1.06$, close to Zipf over several orders of magnitude.

- Later work (e.g., Kondo, Lewis, and Stella 2018) finds the far tail is thinner than Zipf. Still, the striking regularity is that a few very large firms account for much of total output.

Firm-Size Distribution in the U.S.

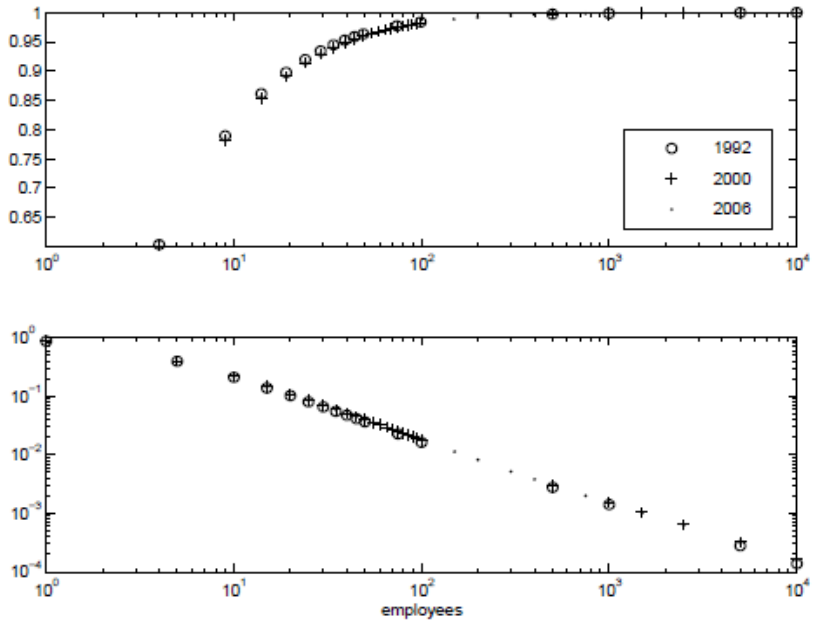


FIGURE 2 The Firm Size Distribution

U.S. firm sizes (employees) in 1992, 2000, and 2006. Top: cumulative distribution. Bottom: the size distribution on log-log scales is close to a straight line, as a power law predicts, and very stable over time. Axtell (2001) estimates $\alpha \approx 1.06$ on the 1997 Census.

Dunne, Roberts, and Samuelson (1989)

- **Data**: U.S. Census of Manufactures, five-year intervals (1963–1982); more than 200,000 plants that entered between 1967 and 1977.
- Plants are grouped into cells by **age** (1–5, 6–10, 11–15 years) and **size** (employees). For each cell they compute:
  - the mean employment growth rate of *surviving* plants;
  - the exit (failure) rate;
  - the mean growth rate of *all* plants, counting an exiting plant as growth of -1.
- Distinguishing the first and third measures matters: a model with selection predicts that survivors look good *because* the bad draws exit.
- They view growth and failure as two outcomes of a single process of market selection. This is exactly the structure of the Jovanovic model that follows.

Dunne, Roberts, and Samuelson (1989), Table I

TABLE I
PLANT GROWTH AND EXIT RATES

	Size (number of employees)					
Age (years)	5–19	20–49	50–99	100–249	>250	Total
a. Mean employment growth rate of successful plants						
1–5	0.606	0.299	0.187	0.132	0.067	0.446
6–10	0.338	0.136	0.066	0.011	−0.011	0.202
11–15	0.310	0.055	−0.006	−0.015	−0.018	0.153
Total	0.519	0.226	0.130	0.077	0.026	0.353
b. Plant exit rates						
1–5	0.412	0.396	0.390	0.327	0.229	0.397
6–10	0.347	0.268	0.281	0.245	0.158	0.303
11–15	0.304	0.206	0.234	0.212	0.131	0.255
Total	0.391	0.347	0.346	0.291	0.191	0.363
c. Mean employment growth rate of all plants						
1–5	−0.056	−0.216	−0.276	−0.238	−0.178	−0.129
6–10	−0.127	−0.169	−0.234	−0.236	−0.167	−0.162
11–15	−0.089	−0.163	−0.239	−0.224	−0.147	−0.141
Total	−0.074	−0.199	−0.261	−0.236	−0.170	−0.138
d. Number of plant-year observations on successful plants/failing plants						
1–5	75,959/53,325	29,938/19,649	13,758/8,794	9,472/4,601	3,281/977	132,408/87,346
6–10	27,409/14,569	15,268/5,584	7,577/2,961	5,829/1,889	2,630/494	58,713/25,947
11–15	7,773/3,400	4,675/1,216	2,198/673	1,568/421	911/137	17,125/5,847
Total	111,141/71,294	49,881/26,449	23,533/12,428	16,869/6,911	6,822/1,608	208,246/118,690

- Panel a: growth of **surviving** plants declines with both size and age.
- Panel b: exit rates decline with both size and age.
- Panel c: once exits count as -1, mean growth is **negative** in every cell; each entry cohort's total employment shrinks over time.

Jovanovic (1982): Overview

- A theory of learning and **"noisy selection"**: efficient firms grow and survive; inefficient firms decline and fail.

- Environment
  - Partial equilibrium: a small industry, constant factor prices, homogeneous product.
  - The path of industry demand is deterministic and known.
  - No aggregate uncertainty, so the output price path $\{p_t\}$ is deterministic and, in equilibrium, self-fulfilling.

- Key friction: **costs are random**. The distribution of efficiency across firms is known to everyone, but no firm knows its own "true" cost. Each firm starts from a common prior and updates as evidence from production comes in.

- Firms and potential entrants choose entry, output, and exit.

- Because the price path is exogenous to any one firm, this is effectively a model of *firm* dynamics embedded in a competitive industry.

Jovanovic (1982): Model

- Cost of producing q_t in period t:

  $$\gamma(q_t)\,\theta(\eta_t), \qquad \eta_t = c + \epsilon_t,$$

  where $\gamma(\cdot)$ is strictly convex, $\theta(\cdot)$ is strictly increasing and positive, c is a firm-specific parameter drawn from a prior $g_0(c)$, and ϵ_t is i.i.d. noise with mean zero (normal in Jovanovic).

- **Timing**: the firm forms its expected cost multiplier $\bar{\theta}_t \equiv E_t[\theta(\eta_t)]$, chooses output, then ϵ_t is drawn; the firm observes its cost, infers η_t, and updates beliefs about c.

- Output solves $\max_q \; p_t q - \gamma(q)\bar{\theta}_t$, so $p_t = \gamma'(q)\bar{\theta}_t$ and

  $$q_t = q\left(\frac{p_t}{\bar{\theta}_t}\right), \qquad q(\cdot) \equiv (\gamma')^{-1}(\cdot) \text{ increasing.}$$

  Firms that believe they are more efficient (low $\bar{\theta}_t$) produce more. Let $\pi(p_t, \bar{\theta}_t)$ denote the maximized profit.

- **Exit**: an exiting firm receives W, the present value of employing its assets in a different activity.
- **Entry**: entrants pay a sunk cost k and start with the prior.

Jovanovic (1982): Learning and Value

- With normal prior and noise, the posterior on c after n periods depends only on the sample mean $\bar{\eta}_n = \frac{1}{n}\sum_{t=1}^{n} \eta_t$ and on n. Since $\theta(\cdot)$ is strictly increasing, $(\bar{\theta}, n)$ is an equivalent **sufficient statistic**.

- Let $H(\cdot \mid \bar{\theta}, n)$ be the distribution of next period's belief. With information $(\bar{\theta}, n)$ and price sequence p:

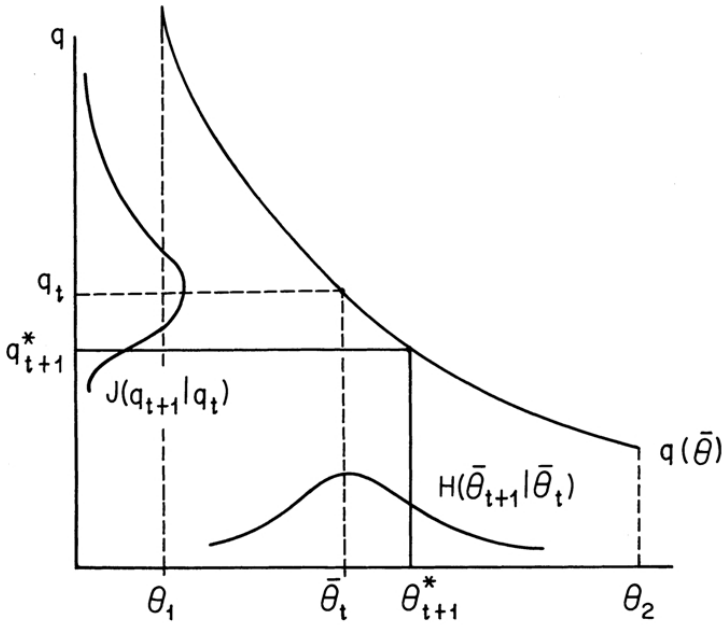
$$V(\bar{\theta}, n, t; p) = \pi(p_t, \bar{\theta}) + \beta \int \max\left[W,\ V(z, n+1, t+1; p)\right] H(dz \mid \bar{\theta}, n).$$

- V is strictly decreasing in $\bar{\theta}$, so $V(\bar{\theta}, n, t; p) = W$ defines a unique **exit threshold** $\bar{\theta}_t^*(n)$: the firm exits if its belief rises above it.

- Let θ_0 be the prior mean. The net value of entry at t is $V(\theta_0, 0, t; p) - k$; free entry requires it to be ≤ 0, with equality whenever entry occurs.

- **Equilibrium**: a price path p such that, given the optimal output, entry, and exit decisions, the market clears in every period.

Jovanovic's Theory in a Graph



- The decreasing curve is output $q(\bar{\theta})$ as a function of beliefs.
- Today the firm believes $\bar{\theta}_t$ and produces q_t.
- Next period's belief is drawn from $H(\bar{\theta}_{t+1}|\bar{\theta}_t)$, centred at $\bar{\theta}_t$.
- If $\bar{\theta}_{t+1} > \bar{\theta}^*_{t+1}$ the firm exits; equivalently, it exits if its output would fall below q^*_{t+1}.
- The induced size distribution $J(q_{t+1}|q_t)$ is truncated at q^*_{t+1}.

Jovanovic (1982): Predictions on Exit

- Beliefs are a **martingale**: $E[\bar{\theta}_{t+1} \mid \bar{\theta}_t] = \bar{\theta}_t$. Firms do not expect to get better or worse; they only learn.
- Size $q_t = q(p_t/\bar{\theta}_t)$ is a random process starting at $q(p_t/\theta_0)$. The shape of $J(q_{t+1} \mid q_t)$ depends on the curvature of $q(\cdot)$ and on $H(\bar{\theta}_{t+1} \mid \bar{\theta}_t)$.
- For each age there is a failure boundary $\bar{\theta}^*_{t+1}$.
- **Exit falls with size, holding age fixed**: a small firm has a high $\bar{\theta}_t$, so it is already close to the boundary and needs only a small bad surprise to cross it.
- **Exit falls with age, holding size fixed**: an older firm has a more precise posterior, so $H(\cdot|\bar{\theta}_t, n)$ has a smaller variance and beliefs move less. (This requires that $\bar{\theta}^*$ not fall too quickly with age.)

Jovanovic (1982): Predictions on Growth

- Let μ_h and σ_h^2 be the mean and variance of the growth rate of **non-failing** firms.
- **Mean growth of survivors falls with size, holding age fixed**. Selection truncates the lower tail of the growth distribution, and the truncation bites harder for small firms, which are closer to the boundary. Survivors among small firms are disproportionately the ones that got good news.
- **Variance of growth falls with age, holding size fixed**: the variance of $H(\bar{\theta}_{t+1} \mid \bar{\theta}_t)$ shrinks as beliefs become more precise.
- These are exactly the patterns in Dunne, Roberts, and Samuelson (1989).

What Jovanovic does not deliver

The model explains *firm* dynamics well, but the industry is not in a stationary state: firms only learn, so eventually all survivors know their type. There is no ongoing turnover in the long run, and the industry structure is essentially exogenous. Hopenhayn (1992) replaces learning about a fixed type with persistent but **changing** productivity, which gives simultaneous entry and exit in a stationary equilibrium.

Competitive Industry Dynamics

- Hopenhayn (1992): a **stationary** model of industry dynamics, with entry, exit, and the firm-size distribution all endogenous.

- "Competitive" is not essential, but firms must act *non-strategically*: each is too small to affect aggregates.
  - Asplund and Nocke (2006) extend it to imperfect (monopolistic) competition. Prediction, confirmed with Swedish hair salons: larger markets have **higher turnover** and **younger** firms.

- A workhorse model that is easy to extend: input-market distortions and misallocation (Hopenhayn and Rogerson 1993; Restuccia and Rogerson 2008), financial frictions, trade (Melitz 2003 is a static cousin).

Key elements

- A continuum of firms, each of measure zero, producing with decreasing returns.
- No aggregate risk: deterministic paths of output and factor prices, taken as given.
- Idiosyncratic risk: each firm's productivity follows a first-order Markov process.
- A sunk cost to enter and a fixed cost to operate each period.

Hopenhayn (1992): Technology

- Time $t = 0, 1, 2, \ldots$; output and input prices p and w taken as given.
- Output y produced with labor n given productivity a:

  $y = a\, f(n), \qquad f$ strictly increasing and strictly concave (DRS).

- Static profits

  $$\pi(a, p, w) \equiv \max_{n} \Big[p\, a\, f(n) - w\, n - k \Big],$$

  where $k > 0$ is the per-period fixed cost of operating.

- Let $n(a, p, w)$ denote optimal employment and $y(a, p, w)$ the associated output.
- DRS is what pins down firm size: with constant returns, the most productive firm would take over the market.

Hopenhayn (1992): Assumptions and Timing

- **Assumptions**
  - $n(\cdot)$, $y(\cdot)$, $\pi(\cdot)$ are all strictly increasing in productivity a.
  - Productivity follows a first-order Markov process with conditional distribution $F(a' \mid a)$, strictly decreasing in a. That is, if $a_1 > a_2$ then $F(\cdot \mid a_1)$ first-order stochastically dominates $F(\cdot \mid a_2)$: a more productive firm today expects to be more productive tomorrow.
  - Entrants pay a sunk cost $k_e > 0$ and draw their initial productivity from a separate distribution $G(a)$.

- **Timing within a period**
  - Incumbents decide to stay or exit; potential entrants decide to enter or not.
  - Incumbents that stay pay k; entrants pay k_e.
  - After paying k or k_e, operating firms learn their productivity draws.

- Exit is a decision about *next* period's (unknown) productivity, so it depends on the firm's current a only through the forecast $F(\cdot|a)$.

The Incumbent's Problem

- Let $z = \{p_t, w_t\}_{t=0}^{\infty}$ denote the sequence of prices a firm takes as given.
- Let $v_t(a, z)$ be the value of an incumbent with current productivity a.
- **Bellman equation**

$$v_t(a, z) = \pi(a, p_t, w_t) + \beta \max\left[0, \int v_{t+1}(a', z)\, dF(a' \mid a)\right].$$

  The outside option is normalized to 0; the fixed cost k is inside π.

- Because π is increasing in a and $F(\cdot|a)$ is FOSD-increasing, v_t is increasing in a. Hence there is an exit threshold $a_t^*(z)$: the firm exits if $a < a_t^*(z)$, where (for interior cases)

$$\int v_{t+1}(a', z)\, dF(a' \mid a_t^*) = 0.$$

- Firms exit not because current profits are negative, but because *expected future* value is negative.

The Entrant's Problem

- Potential entrants are ex ante identical and there is an unlimited pool of them.
- An entrant pays $k_e > 0$, draws its initial productivity from $G(a)$, and starts producing next period.
- Let $m_t \geq 0$ be the mass of entrants. The free-entry condition is

  $$\beta \int v_{t+1}(a, z)\, dG(a) \leq k_e,$$

  with equality whenever $m_t > 0$.
- Free entry pins down the *value* of a new firm, and hence (in a stationary equilibrium) the price p. The *number* of firms then adjusts through m to clear the goods market.

The Aggregate State and Its Law of Motion

- Let $\mu_t(\mathcal{A})$ be the measure (mass) of incumbents with productivity $a \in \mathcal{A}$. It is the state of the industry. It is endogenous and, in general, evolves over time.

- The measure of incumbents with productivity below a' at $t+1$ is

$$\mu_{t+1}([0, a')) = \int F(a' \mid a)\, \mathbb{1}[a \geq a_t^*]\, \mu_t(da) + m_{t+1}\, G(a'), \qquad \text{all } a'.$$

  Survivors transit according to F; entrants arrive according to G (dependence on z suppressed).

- On a grid with N productivity levels this is a **linear** system:

$$\boldsymbol{\mu}_{t+1} = \boldsymbol{\Psi}_t\, \boldsymbol{\mu}_t + m_{t+1}\, \mathrm{g},$$

  where $\boldsymbol{\Psi}_t$ is the $N \times N$ matrix with entries $\Psi_{t,ij} = \Pr(a' = a_i \mid a = a_j)\, \mathbb{1}[a_j \geq a_t^*]$, so it depends on the productivity process and the exit threshold; $\boldsymbol{\mu}$ and g are $N \times 1$ vectors and m is a scalar.

Industry Demand, Supply, and Equilibrium

- Industry demand $D(p)$, exogenous and decreasing.
- Industry supply, endogenous:

$$Y_t = \int y(a, p_t, w_t)\, \mu_t(da).$$

- The market clears when $Y_t = D(p_t)$. Normalize $w_t = 1$.

- Given an initial distribution μ_0, a perfect-foresight equilibrium consists of sequences $\{p_t, m_t, a_t^*, \mu_t\}_{t=0}^{\infty}$ such that
  1. the goods market clears;
  2. incumbents make optimal exit decisions (threshold a_t^*);
  3. there are no further incentives to enter (free entry);
  4. μ_t evolves according to the law of motion.

- We focus on a stationary equilibrium: constants (p^*, m^*, a^*, μ^*) that are a steady state of this dynamical system. The aggregate is constant, yet individual firms keep entering, growing, shrinking, and exiting.

Computing a Stationary Equilibrium (1/2)

- **Step 1.** Guess an output price p_0. For this price, solve the incumbent's dynamic program

  $$v(a, p_0) = \pi(a, p_0) + \beta \max\left[0, \int v(a', p_0)\, dF(a' \mid a)\right].$$

  This is a single-agent contraction. Its solution also gives the exit rule, the $a^*(p_0)$ that solves

  $$\int v(a', p_0)\, dF(a' \mid a^*) = 0.$$

- **Step 2.** Check whether p_0 satisfies free entry:

  $$\beta \int v(a', p_0)\, dG(a') = k_e.$$

  If the left side is too high, entry is too attractive: return to Step 1 with a lower price $p_1 < p_0$. Since v is increasing in p, bisection finds the price p^* that solves the free-entry condition.

Note

Steps 1–2 do not involve demand or the distribution μ. In a stationary equilibrium the price is pinned down by **free entry alone**.

Computing a Stationary Equilibrium (2/2)

- **Step 3.** Guess a mass of entrants m_0 and compute the stationary distribution μ_0 that solves

$$\mu_0([0, a')) = \int_{a \geq a^*(p^*)} F(a' \mid a)\, \mu_0(da) + m_0\, G(a'), \qquad \text{for all } a'.$$

  The right side depends on the price from Step 2 through the exit threshold $a^*(p^*)$.

- **Step 4.** Given μ_0, compute industry supply and check market clearing:

$$Y = \int y(a, p^*)\, \mu_0(da) = D(p^*).$$

  If supply is too low, return to Step 3 with more entrants, $m_1 > m_0$, and continue until m^* clears the market.

Speeding up Step 3–4

The law of motion is linear, so the stationary distribution is **linearly homogeneous in** m:

$$\boldsymbol{\mu} = \boldsymbol{\Psi}\boldsymbol{\mu} + m\,\mathrm{g} \quad \Longrightarrow \quad \boldsymbol{\mu} = m\,(\mathrm{I} - \boldsymbol{\Psi})^{-1}\mathrm{g}.$$

No simulation is needed: build $\boldsymbol{\Psi}$ from $a^*(p^*)$ and invert $(\mathrm{I} - \boldsymbol{\Psi})$ *once*. Supply is then linear in m, so $m^* = D(p^*) / \int y(a, p^*)\, \tilde{\mu}(da)$ with $\tilde{\boldsymbol{\mu}} = (\mathrm{I} - \boldsymbol{\Psi})^{-1}\mathrm{g}$.

Comparative Statics

- **Higher entry cost** k_e
  - Free entry requires a higher expected value of a new firm, so the price p^* rises and incumbents earn more.
  - The exit threshold a^* falls:
    * less selection: incumbents are more profitable, so more of them continue;
    * the average age of firms rises, and more low-productivity firms survive.
  - Entry m^* falls, and so does the turnover rate $m^*/\mu^*(\mathbb{R}_+)$.
  - Aggregate productivity falls: sunk costs **protect inefficient incumbents**.

- **Higher fixed operating cost** k: raises a^* (stronger selection), raises turnover, and raises p^*.

- **Larger demand**: p^* and a^* are pinned down by free entry and do not change. The distribution μ^* and entry m^* scale up proportionally. The model has no market-size effect on firm size or turnover *rates* — one reason Asplund and Nocke add imperfect competition.

Aw, Chung, and Roberts (2003)

- A comparison of Taiwanese and Korean manufacturers: plant-level panels for seven two-digit industries in each country (Taiwan 1981–91, Korea 1983–93).

- **Background**: at the aggregate level Taiwan and Korea are similar export-oriented economies, but their industrial organization differs sharply. Korea's industrial policy favoured large business groups (*chaebol*); Taiwan's manufacturing is dominated by small and medium firms.

- **Goal**
  - interpret relationships among market concentration, producer turnover, and productivity through the lens of a Hopenhayn model;
  - compare these relationships between Taiwan and Korea.

- The next three tables show: (1) concentration, (2) productivity dispersion, (3) turnover. Ask yourself which primitive of the Hopenhayn model could generate all three differences.

Large Firms Account for More Output in Korea

Distribution of Gross Output by Employment Categories

	Taiwan (1986)	South Korea (1988)
Textiles		
5–99 workers	0.235	0.193
100–499 workers	0.400	0.223
500+ workers	0.365	0.553
Apparel		
5–99 workers	0.340	0.285
100–499 workers	0.401	0.383
500+ workers	0.259	0.332
Chemicals		
5–99 workers	0.430	0.140
100–499 workers	0.233	0.361
500+ workers	0.337	0.499
Plastics		
5–99 workers	0.324	0.352
100–499 workers	0.342	0.263
500+ workers	0.334	0.385
Fabricated metals		
5–99 workers	0.640	0.330
100–499 workers	0.272	0.352
500+ workers	0.088	0.317
Electrical machinery/electronics		
5–99 workers	0.184	0.105
100–499 workers	0.262	0.168
500+ workers	0.554	0.727
Transport equipment		
5–99 workers	0.225	0.070
100–499 workers	0.237	0.114
500+ workers	0.538	0.816

Share of gross output by plant-size class. In most industries plants with 500+ workers produce a much larger share in Korea (e.g., transport equipment: 82% vs. 54%).

Productivity Dispersion Is Higher in Korea

Cross-sectional Productivity Dispersion (average values over three census years)

Industry	Taiwan (1981–91)	South Korea (1983–93)
Textiles		
Standard Deviation	0.266	0.380
Range (10th–90th Percentile)	0.635	0.893
Mean *TFP* Growth	0.159	0.095
Apparel		
Standard Deviation	0.273	0.376
Range (10th–90th Percentile)	0.643	0.910
Mean *TFP* Growth	0.039	0.217
Chemicals		
Standard Deviation	0.193	0.013
Range (10th–90th Percentile)	0.255	0.400
Mean *TFP* Growth	0.576	0.936
Plastics		
Standard Deviation	0.237	0.314
Range (10th–90th Percentile)	0.553	0.713
Mean *TFP* Growth	0.119	0.120
Fabricated Metals		
Standard Deviation	0.244	0.330
Range (10th–90th Percentile)	0.575	0.791
Mean *TFP* Growth	0.052	0.083
Electrical Machinery/Electronics		
Standard Deviation	0.236	0.326
Range (10th–90th Percentile)	0.546	0.756
Mean *TFP* Growth	0.173	0.061
Transport Equipment		
Standard Deviation	0.240	0.310
Range (10th–90th Percentile)	0.550	0.722
Mean *TFP* Growth	−0.020	0.115

Less Turnover in Korea

Industry Output Volatility Rates

Industry	Taiwan		South Korea	
	1981–6	1986–91	1983–8	1988–93
Textiles	0.847	0.829	0.335	0.656
Apparel	1.00	0.881	0.750	1.10
Chemicals	0.540	0.655	0.265	0.158
Plastics	0.719	0.886	0.618	0.558
Fabricated Metals	0.788	0.918	0.786	0.858
Electrical Machinery/Electronics	0.602	0.640	0.370	0.454
Transport Equipment	0.451	0.510	0.432	0.194

Volatility is defined as (entry rate + exit rate) − |entry rate − exit rate| = 2 Min(entry rate, exit rate). Entry and exit rates are measured using the value of output accounted for by the entering and exiting firms. See Dunne and Roberts (1991) or Roberts (1996) for details.

Volatility $=$ (entry rate $+$ exit rate) $-$ $|$entry rate $-$ exit rate$|$ $= 2\,\min($entry rate, exit rate$)$: "excess" turnover, net of the industry's expansion or contraction. Rates are weighted by output.

Aw, Chung, and Roberts (2003): Summary

- Relative to Taiwan, Korean industries are characterized by
  - more market concentration;
  - more cross-sectional productivity dispersion;
  - less producer turnover;
  - more low-productivity producers operating;
  - more output attributable to high-productivity producers;
  - larger productivity differences between surviving and failing producers.

- This is what a Hopenhayn model predicts if the sunk entry cost k_e is **higher in Korea**:
  - a higher k_e discourages entry;
  - a higher k_e protects incumbents;

  $\Rightarrow$ the productivity threshold a^* falls, the selection effect on existing producers weakens, and more incumbents are low-productivity firms.

Interpretation

The model turns three separate facts into one diagnosis: barriers to entry and exit that insulate inefficient producers from market pressure.

From Competitive to Strategic Industry Dynamics

- In Hopenhayn, each firm is measure zero, the law of large numbers makes the aggregate state μ_t deterministic, and each firm solves a **single-agent** problem against a price path.

- Many IO questions are about **concentrated** industries, where each firm's actions move the state that its rivals care about:
  - preemptive investment and capacity races;
  - R&D races and innovation;
  - learning-by-doing and predatory pricing;
  - dynamic effects of mergers and of entry regulation.

- With finitely many firms:
  - the industry state is **random**, even without aggregate shocks;
  - each firm must forecast how its rivals' states evolve, and those depend on rivals' strategies, which depend on what they expect of *it*.

- The equilibrium concept is a **Markov perfect equilibrium** (Maskin and Tirole 1988).

Markov Perfect Equilibrium

Basics of a Markov perfect equilibrium (MPE):

- A refinement of subgame-perfect Nash equilibrium for dynamic games: strategies may depend on history only through the current **payoff-relevant state**.
- Multiple agents interact non-cooperatively, repeatedly over time.
- There is a common state vector whose path both influences and is influenced by the agents' decisions.
- The transition of the state that each agent faces depends on the decisions of the other agents, and the agent takes that into account.
- Each agent solves a dynamic program that includes this law of motion.
- An MPE prevails if no agent wishes to revise its policy, taking as given the policies of all others.

Why Markov?

It rules out history-dependent (e.g., collusive trigger) strategies and turns the problem into a computable fixed point in *functions of the state*.

Warm-up: A Dynamic Duopoly

A linear-quadratic example (Ljungqvist and Sargent, ch. 7).

- Market demand: $p = a_0 - a_1(q_1 + q_2)$.
- Firm i chooses next period's output $\hat{q}_i$ and pays a quadratic adjustment cost:

$$\pi_i(q_i, q_{-i}, \hat{q}_i) = a_0 q_i - a_1 q_i^2 - a_1 q_i q_{-i} - \gamma(\hat{q}_i - q_i)^2.$$

- Given the rival's policy $\hat{q}_{-i} = f_{-i}(q_{-i}, q_i)$, firm i's Bellman equation is

$$v_i(q_i, q_{-i}) = \max_{\hat{q}_i} \left\{ \pi_i(q_i, q_{-i}, \hat{q}_i) + \beta\, v_i\big(\hat{q}_i, f_{-i}(q_{-i}, q_i)\big) \right\}.$$

- Mapping into the general concepts
  - state: $s = (q_i, q_{-i})$;
  - (deterministic) law of motion: $s' = (\hat{q}_i,\ f_{-i}(s))$, which depends on the rival's policy.
- An MPE is a pair of value functions (v_1, v_2) and policies (f_1, f_2) such that each f_i solves firm i's problem given f_{-i}.
- Here the policies are linear and can be found by iterating on two coupled Riccati equations. Adjustment costs make output a **state** and create a strategic incentive to commit to a higher output.

Ericson and Pakes (1995): Motivation

- Motivated by the empirical facts above (Dunne–Roberts–Samuelson, Davis–Haltiwanger):
  1. heterogeneity among firms within narrowly defined markets;
  2. simultaneous entry and exit;
  3. uncertainty at both the firm and the market level.

- ... while retaining strategic interaction: a **stochastic dynamic game**.
  1. Earlier theory showed that stylized dynamic games can generate almost any pattern of outcomes; the answer depends on details.
  2. The MPE refinement is very useful for applied work.
  3. So EP propose a **computational** framework that aims for realistic complexity: write down primitives, compute the equilibrium, simulate.

- Outputs of the framework: value and policy functions for incumbents and potential entrants in every industry state, and the implied Markov process for the industry structure.

- Reference implementation and survey: Doraszelski and Pakes (2007), *Handbook of IO*, vol. 3.

Ericson–Pakes: Environment

- Discrete time, infinite horizon. Incumbent firms and a pool of potential entrants.

- **Static–dynamic breakdown**: the product-market game (Cournot, differentiated Bertrand, ...) is static, so per-period profits $\pi(\omega_i, \omega_{-i})$ can be computed "offline" for every state before solving the dynamic game.

- **Timing each period**
  1. Incumbents observe their private scrap value and decide whether to exit; those that stay choose investment x_i.
  2. Potential entrants observe their private entry cost and decide whether to enter.
  3. Product-market competition takes place; profits are realized.
  4. Investment outcomes and industry shocks are realized; entry and exit take effect; the new state is determined.

- **Individual state** $\omega_i \in \Omega = \{1, 2, \ldots, \bar{\omega}\}$: a summary of firm i's competitiveness (quality, capacity, cost). The industry state is the list of incumbents' states $(\omega_1, \ldots, \omega_n)$, $n \leq \bar{n}$; we write ω_{-i} for the states of i's rivals.

Ericson–Pakes: Symmetry, Anonymity, and the State Space

- The literature imposes **symmetry** (all firms share the same primitives and policy functions) and **anonymity** (a firm cares about how many rivals are in each state, not who they are).

- Then the industry state can be written as a **count vector** (recall the measure μ in Hopenhayn):

$$s = (s_1, \ldots, s_{\bar{\omega}}), \qquad s_\omega = \#\{i : \omega_i = \omega\}, \qquad \sum_\omega s_\omega \leq \bar{n}.$$

  Example with $\bar{\omega} = 4$: the ordered list $(3, 1, 1)$ and the list $(1, 3, 1)$ are the same state $s = (2, 0, 1, 0)$.

- **Size of the state space** S:
  - ordered lists: $(\bar{\omega} + 1)^{\bar{n}}$ (each slot is empty or in one of $\bar{\omega}$ states) — exponential in $\bar{n}$;
  - count vectors: $\binom{\bar{n}+\bar{\omega}}{\bar{n}}$ — polynomial in $\bar{n}$ for fixed $\bar{\omega}$.
- Example: $\bar{n} = 6$, $\bar{\omega} = 20$: $21^6 \approx 8.6 \times 10^7$ ordered states versus $\binom{26}{6} = 230{,}230$ count vectors.

- A firm's problem is then indexed by (ω_i, s_{-i}): its own state and the counts of its rivals. To save notation we keep writing (ω_i, ω_{-i}).

Ericson–Pakes: State Transitions

- EP want both **firm-specific** and **industry-wide** uncertainty. Each incumbent's state evolves as

  $$\omega_i' = \omega_i + \nu_i - \eta.$$

- **Own investment outcome** $\nu_i \in \{0, 1\}$, stochastically increasing in investment x_i:

  $$\Pr(\nu_i = 1 \mid x_i) = \frac{\alpha x_i}{1 + \alpha x_i}, \qquad \Pr(\nu_i = 0 \mid x_i) = \frac{1}{1 + \alpha x_i}.$$

  The success probability is concave in x_i and approaches 1 only as $x_i \to \infty$.

- **Industry shock** $\eta \in \{0, 1\}$, common to all firms: $\eta = 1$ with probability δ. It captures, e.g., an improving outside good, so standing still means falling behind.

- This is a "step-by-step" innovation technology; richer technologies are easy to accommodate.

- Why a common shock? EP think regulation, new technological opportunities, etc. make incumbents' profits *positively* correlated. Some papers instead use a **firm-specific** depreciation shock η_i, which implies a different (often negative, business-stealing) correlation of profits.

The Incumbent's Problem (1/2): Bellman Equation

- An incumbent observes its scrap value ϕ, drawn i.i.d. from F^{ϕ} and privately known. Its value is

$$V(\omega_i, \omega_{-i}, \phi) = \pi(\omega_i, \omega_{-i}) + \max\left\{\phi,\ \max_{x_i \geq 0}\left[-x_i + \beta\, E[\bar{V}(\omega_i', \omega_{-i}') \,|\, \omega_i, \omega_{-i}, x_i]\right]\right\},$$

  where $\bar{V}(\omega) \equiv \int V(\omega, \phi)\, dF^{\phi}(\phi)$ integrates out next period's scrap value.

- Separate the **own, endogenous** uncertainty (ν_i) from the rest. Define the expected value conditional on the own outcome ν:

$$W(\nu \,|\, \omega_i, \omega_{-i}) = \sum_{\eta} p(\eta) \sum_{\omega_{-i}'} \bar{V}(\omega_i + \nu - \eta,\ \omega_{-i}')\ \mathsf{q}(\omega_{-i}' \,|\, \omega_i, \omega_{-i}, \eta).$$

  Then $E[\bar{V}(\omega_i', \omega_{-i}') \,|\, \omega, x_i] = \sum_{\nu} W(\nu \,|\, \omega)\, p(\nu \,|\, x_i)$.

- $\mathsf{q}(\omega_{-i}' \,|\, \omega_i, \omega_{-i}, \eta)$ is the perceived transition kernel: firm i's beliefs about where its rivals (including entrants and exiters) will be next period. It is an **equilibrium object**: it is generated by the rivals' policies.

The Incumbent's Problem (2/2): Investment and Exit

- **Investment.** Given W, investment solves a one-dimensional problem. With $\Delta W \equiv W(1|\omega) - W(0|\omega)$,

  $$\max_{x \geq 0} \; -x + \beta \Big[W(0|\omega) + \Delta W \, \frac{\alpha x}{1 + \alpha x} \Big].$$

  The first-order condition $\beta \, \alpha \, \Delta W / (1 + \alpha x)^2 = 1$ gives the closed form

  $$x^*(\omega) = \max \Big\{ 0, \; \frac{\sqrt{\alpha \beta \, \Delta W} - 1}{\alpha} \Big\}.$$

  Firms invest more when the gain from success ΔW is larger.

- **Exit.** Let $C(\omega) \equiv -x^*(\omega) + \beta \sum_\nu W(\nu|\omega) \, p(\nu|x^*(\omega))$ be the continuation value. The firm stays iff $\phi \leq C(\omega)$, so the probability of staying is

  $$r(\omega) = F^\phi \big[C(\omega) \big].$$

- **Integrated value.**

  $$\bar{V}(\omega) = \pi(\omega) + \big(1 - r(\omega)\big) \, E \big[\phi \,|\, \phi > C(\omega) \big] + r(\omega) \, C(\omega).$$

- Randomizing the scrap value makes exit a smooth function of C, which is what guarantees existence of an equilibrium in pure (cutoff) strategies (Doraszelski and Satterthwaite 2010).

The Potential Entrant's Problem

- Each period there are $\bar{\epsilon}$ potential entrants (or $\bar{n} - n$). An entrant draws a private entry cost $\phi^e \sim F^e$, decides whether to enter, and if so appears next period at an initial state ω^e.

- The value of entering in industry state ω (before paying ϕ^e) is

$$V^e(\omega) = \beta \sum_{\eta} p(\eta) \sum_{\omega'} \bar{V}\left(\omega^e, \omega'\right) \, \mathrm{q}^e(\omega' \,|\, \omega, \eta).$$

- The entrant enters iff $\phi^e \leq V^e(\omega)$, so the probability of entry is

$$r^e(\omega) = F^e\left[V^e(\omega)\right].$$

- The entrant forms beliefs over a slightly different object: q^e is the distribution of *all* incumbents' states next period (it is not yet one of them).

- Extensions: entrants can invest before entering (as in the original EP and Pakes–McGuire), or draw a random initial state.

Equilibrium

- A (symmetric) **Markov perfect equilibrium** is a set of functions $\{\bar{V}, x, r, r^e\}$ on the state space and perceived kernels $\{\mathsf{q}, \mathsf{q}^e\}$ such that:
  1. **Optimality**: each incumbent and potential entrant chooses optimal strategies *given* its perceptions $\mathsf{q}(\cdot|\cdot)$ and $\mathsf{q}^e(\cdot|\cdot)$;
  2. **Consistency**: these perceptions are the transition probabilities actually generated by the competitors' equilibrium policies $\{x, r, r^e\}$.

- A rigorous treatment is in Doraszelski and Satterthwaite (2010):
  - With privately drawn scrap values and entry costs, an equilibrium in **cutoff** entry/exit strategies always exists. Without them, existence may require mixed entry/exit strategies, contrary to the original EP claim.
  - A condition on primitives ensures investment is in pure strategies (the single-agent investment problem must be well behaved, as in the closed form above).
  - A symmetric equilibrium exists under symmetric primitives.

Properties of the Equilibrium

- **Finite state space.** Since ω_i is bounded and per-firm profits fall as rivals are added, no more than $\bar{n}$ firms are ever active. So the state space S is finite and, in principle, computable.

- **A Markov chain for the industry.** Any equilibrium kernel defines a time-homogeneous Markov process for the industry state s.

- **Recurrent class.** In finite time the industry wanders into a recurrent class $R \subseteq S$ and never leaves. EP show conditions under which there is only one such class. R is often much smaller than S. This matters for computation (Pakes–McGuire 2001) and for estimation, if we are willing to assume that the observed data are already in R.

- **Multiplicity.** In general there are **multiple equilibria**. Doraszelski and Pakes (2007) give examples from static competition, entry/exit, and investment; Besanko et al. (2010) map out the full set in one model (Part VI). As with static entry games, this complicates estimation and counterfactuals.

Pakes and McGuire (1994): Overview

- The most transparent algorithm: a **Gauss–Jacobi iteration** on the value and policy functions over the whole state space.

- **In memory**, for every industry state $\omega \in S$ (count vector), store
  - the incumbent's value $\bar{V}(\omega_i, \omega_{-i})$, investment $x(\cdot)$, and probability of staying $r(\cdot)$ for each occupied own state ω_i;
  - the potential entrant's value $V^e(\omega)$ and probability of entry $r^e(\omega)$.

- **Initial values** (fairly arbitrary):

$$\bar{V}^0(\omega_i, \omega_{-i}) = \frac{\pi(\omega_i, \omega_{-i})}{1-\beta}, \qquad r^0(\cdot) = 1, \qquad x^0(\cdot) = 0, \qquad V^{e,0}(\cdot) = r^{e,0}(\cdot) = 0.$$

- Notation: write $f(\omega)_i \equiv f(\omega_i, \omega_{-i})$ for firm i's value of a function in state ω.

- Each iteration l takes the values and policies of iteration $l-1$ as given and, **state by state**, performs the four steps on the next slides.

Pakes–McGuire Step 1: Rivals' Outcome Probabilities

For a given state ω and each incumbent j, use the iteration-$(l-1)$ policies to compute the probability of each outcome (before the common shock η):

Outcome for firm j	Next state (before η)	Probability p_j^{l-1}
exits	—	$1 - r^{l-1}(\omega)_j$
stays, investment succeeds	$\omega_j + 1$	$r^{l-1}(\omega)_j \; p\big(1 \mid x^{l-1}(\omega)_j\big)$
stays, investment fails	ω_j	$r^{l-1}(\omega)_j \; \big[1 - p\big(1 \mid x^{l-1}(\omega)_j\big)\big]$

- Then subtract the common shock: with probability δ every surviving firm moves down one step.
- **Entry.** With $\bar{\epsilon}$ potential entrants each entering with probability $r^{e,l-1}(\omega)$, the number of entrants ϵ' is binomial:

$$\Pr^{l-1}(\epsilon' \mid \omega) = \binom{\bar{\epsilon}}{\epsilon'} r^{e,l-1}(\omega)^{\epsilon'} \left[1 - r^{e,l-1}(\omega)\right]^{\bar{\epsilon}-\epsilon'}.$$

  Entrants appear at ω^e.

Pakes–McGuire Step 2: Kernel and Continuation Values

- Firms' outcomes are independent given the state and η. So the **perceived transition kernel** of incumbent i is a product over its rivals and the entrants:

$$\mathsf{q}^{l}(\omega'_{-i} \,|\, \omega, \eta) = \sum_{\substack{\text{rival outcomes and } \epsilon' \\ \text{that lead to } \omega'_{-i}}} \Big[\prod_{j \neq i} p_j^{l-1}\Big] \; \Pr^{l-1}(\epsilon' \,|\, \omega).$$

- Update the **expected continuation value** given own outcome $\nu \in \{0, 1\}$:

$$W^{l}(\nu \,|\, \omega)_i = \sum_{\eta} p(\eta) \sum_{\omega'_{-i}} \bar{V}^{l-1}\big(\omega_i + \nu - \eta,\ \omega'_{-i}\big) \; \mathsf{q}^{l}(\omega'_{-i} \,|\, \omega, \eta).$$

- This step is where the time goes: with K possible outcomes per rival and n incumbents, the inner sum has on the order of K^{n-1} terms, *per firm, per state*.

- Note what is *not* done: we never solve each firm's DP to convergence given the others' policies. Values and policies are updated together, one step at a time.

Pakes–McGuire Step 3–4: Update Policies and Values

- **Investment**: use the Kuhn–Tucker condition (closed form above) with $\Delta W^l = W^l(1|\omega)_i - W^l(0|\omega)_i$:

  $$x^l(\omega)_i = \max\left\{0,\ \left(\sqrt{\alpha\beta\,\Delta W^l} - 1\right)/\alpha\right\}.$$

- **Exit**: with $C^l(\omega)_i = -x^l(\omega)_i + \beta \sum_\nu W^l(\nu|\omega)_i\, p(\nu|x^l(\omega)_i)$,

  $$r^l(\omega)_i = F^\phi\left[C^l(\omega)_i\right].$$

- **Value**:

  $$\bar{V}^l(\omega)_i = \pi(\omega)_i + \left(1 - r^l(\omega)_i\right) E\left[\phi \,|\, \phi > C^l(\omega)_i\right] + r^l(\omega)_i\, C^l(\omega)_i.$$

  (With a deterministic scrap value, the middle term is just $(1 - r^l)\phi$.)

- **Entry**: compute $V^{e,l}(\omega)$ from $\bar{V}^{l-1}$ and $\mathrm{q}^{e,l}$, and set $r^{e,l}(\omega) = F^e\left[V^{e,l}(\omega)\right]$.

- Repeat for every $\omega \in S$; then move to iteration $l + 1$.

Pakes–McGuire: Convergence and Practical Issues

- **Stopping rule**: stop when $\max_\omega \left|\bar{V}^l(\omega) - \bar{V}^{l-1}(\omega)\right|$ (and the change in policies) is below a tolerance.

- **Gauss–Jacobi vs. Gauss–Seidel**: Jacobi uses only iteration-$(l-1)$ values; Gauss–Seidel uses the values already updated in iteration l as soon as they are available, which usually speeds things up.

- **No convergence guarantee**. Unlike a single-agent Bellman operator, the best-response mapping in a game is not a contraction. In practice:
  - damp the updates, $\bar{V}^l \leftarrow \lambda \bar{V}^l + (1-\lambda)\bar{V}^{l-1}$ with $\lambda < 1$;
  - try different starting values.

- **Which equilibrium?** If the algorithm converges, it finds *an* equilibrium. Different starting values or orderings may find different ones, and some equilibria are unstable under this iteration and can never be found by it. Homotopy methods (Besanko et al. 2010) trace out many equilibria systematically.

The Curse of Dimensionality

- Computing time $\approx$ (number of states) $\times$ (time per state) $\times$ (number of iterations).
- **Number of states.** Even with symmetry and anonymity, $\#S = \binom{\bar{n}+\bar{\omega}}{\bar{n}}$, which grows quickly in both $\bar{n}$ and $\bar{\omega}$. With $\bar{n} = 6$ and $\bar{\omega} = 20$, every function must be stored on about 230,000 states.
- **Time per state.** Each of up to n incumbents needs an expectation over its rivals' outcomes. With 3 outcomes per rival (exit, up, same) and the binary common shock, that is $2 \cdot 3^{n-1} = 486$ terms per firm when $n = 6$.
  - One sweep: $\approx 230{,}000 \times 6 \times 486 \approx 7 \times 10^8$ evaluations, before entry.
- **Number of iterations** typically grows with the size of the problem as well.
- This is why applied work with Pakes–McGuire rarely goes beyond a handful of firms with a few states each.

Alleviating the Computational Burden

1. **Continuous time** (Doraszelski and Judd 2012). In a short interval at most one firm changes its state, so the expectation sums over $O(nK)$ terms rather than K^n. Speed-ups of orders of magnitude.

2. **Functional approximation**: approximate the value function by a low-dimensional parametric function of the state (e.g., approximate linear programming, Farias, Saure, and Weintraub 2012).

3. **Stochastic approximation**: Pakes and McGuire (2001). Never compute the expectation and never visit states outside the recurrent class.

4. **Alternative equilibrium concepts**: firms condition on less information. Oblivious equilibrium (Weintraub, Benkard, and Van Roy 2008) and moment-based Markov equilibrium (Ifrach and Weintraub 2017).

We go over 3 and 4.

Pakes and McGuire (2001): A Stochastic Algorithm

- Two departures from the 1994 algorithm, each attacking one source of the curse:
  - **Asynchronous**: each iteration updates values and policies at a *single* industry state, the one the simulated industry is currently in. Only states in the recurrent class R are visited repeatedly.
  - **Monte Carlo integration**: the continuation value W is never computed as a sum over K^{n-1} terms. It is estimated by averaging the values of *past simulated outcomes*.

- If taken literally as behaviour, firms "learn" continuation values from experience; this links the algorithm to the learning literature (and to reinforcement learning).

- It trades precision for memory and speed. It works best when
  1. the number of state variables is large;
  2. the recurrent class is small relative to the whole state space;
  3. the transition kernel is complicated to compute.
- It needs a more careful stopping rule, since estimates are noisy.

Pakes and McGuire (2001): The Basic Idea

- Consider any set of numbers $W(\nu \mid \omega_i, \omega_{-i})$, one for each own outcome and location. These need not be equilibrium objects.

- Given W, each firm's problem at a state is a **static** optimization: we can compute $V(\omega_i, \omega_{-i} \mid W)$, the investment policy $x(\omega_i, \omega_{-i} \mid W)$, and the exit policy $r(\omega_i, \omega_{-i} \mid W)$ without any integration.

- The equilibrium W is a fixed point:

$$W(\nu \mid \omega_i, \omega_{-i}) = \sum_{\eta} p(\eta) \sum_{\omega'_{-i}} V\left(\omega_i + \nu - \eta,\ \omega'_{-i} \,\middle|\, W\right)\ \mathrm{q}(\omega'_{-i} \mid \omega_i, \omega_{-i}, W).$$

  Both the values on the right and the kernel q depend on W through the implied policies.

- So we are searching for a set of numbers $W(\cdot)$ that satisfy this mapping. The right-hand side is an **expectation**, which we can estimate by sampling from q instead of summing over it.

- How do agents update their beliefs about W? By accumulating information from the outcomes they experience over time.

Pakes and McGuire (2001): One Iteration

Let k index iterations, s^k the current industry state, and W^k the current estimates.

1. **Policies at the current state.** For each incumbent at s^k, solve

$$V(\omega_i, s^k \mid W^k) = \pi(\omega_i, s^k) + \max\left\{\phi,\ \sup_x \left[-x + \beta \sum_\nu W^k(\nu \mid \omega_i, s^k)\, p(\nu|x)\right]\right\},$$

   giving $x(\omega_i, s^k|W^k)$ and the exit decision; likewise the entry decision.

2. **Simulate the next state.** Draw each firm's outcome $\nu_j \sim p(\cdot|x_j)$, the common shock η^{k+1}, and entry and exit. This gives s^{k+1}.

3. **Update the estimates at s^k only** (all other cells are left unchanged), for each firm and *both* values of ν:

$$W^{k+1}(\nu \mid \omega_i, s^k) = W^k(\nu \mid \omega_i, s^k) + \alpha_k \left[V\left(\omega_i + \nu - \eta^{k+1},\ s_{-i}^{k+1} \mid W^k\right) - W^k(\nu \mid \omega_i, s^k)\right].$$

Pakes and McGuire (2001): Reading the Update

- $V(\omega_i + \nu - \eta^{k+1}, s_{-i}^{k+1} \mid W^k)$ is the value at *tomorrow's* simulated state, computed with the current estimates there (a one-step look-ahead, as in temporal-difference learning).
- It combines the **realized** moves of the rivals and the common shock, $(s_{-i}^{k+1}, \eta^{k+1})$, with a **hypothetical** own outcome ν. So both $W(1|\cdot)$ and $W(0|\cdot)$ are updated every time a state is visited.
- With step size $\alpha_k = 1/h_k(s^k)$, where $h_k(s^k)$ is the number of past visits to s^k, $W^k(\nu|\omega_i, s)$ is the **sample average** of the realized continuation values from that state. A draw from q replaces the sum over K^{n-1} terms.
- Memory: only visited states are stored (e.g., in a hash table), so memory scales with $\#R$, not $\#S$.
- **Stopping rule.** Fix W, simulate a long path, and compare W with the sample average of the realized values $V(\cdot \mid W)$, weighting states by their visit frequency in R. If they agree, W is (approximately) an equilibrium on the recurrent class. Fershtman and Pakes (2012) formalize this as an "experience-based equilibrium".

Approximating Markov Perfect Equilibrium

- MPE requires each firm to form expectations over *all* of its rivals' strategies.
  - Heavy computational burden in the transition kernel.
  - Firms may not be so well informed in realistic settings.

- If the industry is **not concentrated**:
  - a "monopolistically competitive" assumption is reasonable and the Hopenhayn intuition applies;
  - an integer number of firms + Hopenhayn = oblivious equilibrium (OE; Weintraub, Benkard, and Van Roy 2008). Each firm conditions only on its own state and plays against the **long-run average industry state**.

- With finitely many firms, the actual industry state is still a random Markov process; the "average industry state" is its expectation under the long-run (invariant) distribution implied by the OE strategies.

- As the number of firms $\to \infty$, OE $\to$ Hopenhayn. WBV show OE approximates MPE well in large markets, provided the equilibrium firm-size distribution satisfies a **light-tail** condition: no firm is large enough to move aggregates.

Moment-Based Markov Equilibrium: Motivation

- Many industries combine **high concentration** with a **large number of firms**: a few dominant firms and a competitive fringe (national chains and local stores; large airlines and regional carriers).

- OE fails here: dominant firms are not "light-tailed", and their states matter for everyone. MPE is infeasible with many fringe firms.

- Ifrach and Weintraub (2017) introduce a new equilibrium concept, the moment-based Markov equilibrium (MME). Firms choose strategies based on
  1. their own state;
  2. the detailed state of the dominant firms;
  3. a few aggregate moments of the distribution of fringe firms.
- They introduce a Markov *perceived* transition kernel to approximate the dynamics of the (non-Markov) moment-based state.

- Behaviourally appealing (managers track big rivals closely and the fringe roughly) and computationally tractable. Similar in spirit to Krusell and Smith (1998) in macro.

MME: Notation

- Discrete time t; individual state $x_{it} \in \mathcal{X}$ (note: x now denotes the *state*, following the paper).
- Industry state $\bar{s}_t$, symmetric and anonymous (a count vector); $n_t \leq N$ incumbent firms.
- Investment ι_{it} with cost $c(\iota, x)$; endogenous transitions

$$Q[x' \mid x, \iota] = \Pr[x_{i,t+1} = x' \mid x_{it} = x,\ \iota_{it} = \iota].$$

- Incumbents draw a private i.i.d. sell-off value ϕ_{it}.
- There are $N - n_t$ potential entrants, each with a private i.i.d. entry cost κ_{it}; entrants start at $x^e \in \mathcal{X}$.
- Aggregate industry shock z_t.
- Single-period payoff $\pi(x_{it}, \bar{s}_t, z_t)$.

MME: The Benchmark MPE

- State space $\mathcal{S} = \left\{(\bar{s}, z) : \sum_{x \in \mathcal{X}} \bar{s}(x) \leq N\right\}$; write $s_t = (\bar{s}_t, z_t)$.
- Incumbent strategies $\mu = (\iota, \rho)$: invest $\iota(x, s)$, and exit if $\phi_{it} > \rho(x_{it}, s_t)$.
- Entrant strategy λ: enter if $\kappa_{it} < \lambda(s_t)$.

- Value of an incumbent using μ' when all rivals use (μ, λ), with τ_i its exit time:

$$V(x, s \,|\, \mu', \mu, \lambda) = E_{\mu', \mu, \lambda}\left[\sum_{k=t}^{\tau_i} \beta^{k-t}\big(\pi(x_{ik}, s_k) - c(\iota_{ik}, x_{ik})\big) + \beta^{\tau_i - t}\phi_{i,\tau_i} \;\middle|\; x_{it} = x,\ s_t = s\right].$$

- An MPE is (μ, λ) such that
  1. $\sup_{\mu'} V(x, s \,|\, \mu', \mu, \lambda) = V(x, s \,|\, \mu, \mu, \lambda)$ for all (x, s): no profitable deviation;
  2. $\lambda(s) = \beta\, E_{\mu, \lambda}\left[V(x^e, s' \,|\, \mu, \mu, \lambda) \,|\, s\right]$ for all s: the entry cutoff equals the expected value of entry.

MME: Dominant and Fringe Firms

- Partition individual states into fringe and dominant: $\mathcal{X}_f$ and $\mathcal{X}_d$.
- Fringe states: $\mathcal{S}_f = \{f : \sum_{x \in \mathcal{X}_f} f(x) \leq N\}$, with f a particular fringe distribution.
- Dominant firms' state $d \in \mathcal{S}_d$, with $\bar{D}$ the maximum number of dominant firms.

- **Moment-based strategies**
  - A moment function, e.g., $\theta(f) = \sum_{x \in \mathcal{X}_f} x^\alpha f(x)$.
  - Strategies depend only on $(x, d, \theta(f), z)$.
  - The moment-based industry state is $\hat{s} = (\theta, d, z)$.

- This is attractive when profits depend on the fringe only through such a moment, e.g., logit demand where fringe firms act as price takers: their effect on dominant firms runs through one "inclusive value"-type sum.

MME: Why the Moment Is Not Markov

- θ may be enough to compute **today's** profits, but not enough to predict **tomorrow's** θ.

- Example (first moment): $\theta = 10$ is consistent with one fringe firm in state 10 *and* with ten fringe firms in state 1. The two industries evolve very differently.

- So the process $\{\theta_t\}$ is generally **not Markov**, even though the underlying process for the full state is. Information is lost in aggregation, and past values of θ help predict future ones.

- The same issue arises with the "inclusive value" in dynamic demand (Gowrisankaran–Rysman) and with Krusell–Smith in macro.

- Solution: postulate a **perceived** Markov kernel $\hat{P}_{\mu,\lambda}[\theta' \mid \hat{s}]$ that firms use as if it were true, and require it to be consistent with the frequencies actually observed in equilibrium (as in Fershtman–Pakes and Krusell–Smith).

MME: The Perceived Transition Kernel

- θ also does not contain enough information to say when a fringe firm becomes dominant. More structure is needed:
  - at most one fringe firm becomes dominant per period;
  - following the Pakes–McGuire (2001) idea, use long-run empirical frequencies:

$$\hat{P}_{\mu,\lambda}[f \to d \mid \hat{s}] = \limsup_{T\to\infty} \frac{\sum_{t=1}^{T} \mathbb{1}\{\hat{s}_t = \hat{s},\ f \to d\}}{\sum_{t=1}^{T} \mathbb{1}\{\hat{s}_t = \hat{s}\}};$$

  - if $\bar{D}$ is reached, this probability is zero.
- No approximation is needed for $d \to f$, since dominant firms are tracked individually.

- The overall perceived transition kernel has two parts:
  - $P_{\mu',\mu,\lambda}[x', d', z' \mid x, \hat{s}]$: own, dominant firms', and aggregate states, **exact**, as in a standard MPE;
  - $\hat{P}_{\mu,\lambda}[\theta' \mid \hat{s}] \times \hat{P}_{\mu,\lambda}[f \to d \mid \hat{s}]$: the fringe, **approximated**.
- A fringe firm ignores its own effect on the evolution of the moments.

Moment-Based Markov Equilibrium

- The *perceived* value of an incumbent using μ' when rivals use (μ, λ):

$$\hat{V}(x, \hat{s} \mid \mu', \mu, \lambda) = E_{\mu', \mu, \lambda}\Big[\sum_{k=t}^{\tau_i} \beta^{k-t} \big(\pi(x_{ik}, \hat{s}_k) - c(\iota_{ik}, x_{ik})\big) + \beta^{\tau_i - t} \phi_{i,\tau_i} \;\Big|\; x_{it} = x,\ \hat{s}_t = \hat{s} \Big].$$

  The expectation uses the **perceived** kernel $\hat{P}_{\mu', \mu, \lambda}$ from the previous slide.

- An MME is a pair of moment-based strategies (μ, λ) such that
  1. $\sup_{\mu'} \hat{V}(x, \hat{s} \mid \mu', \mu, \lambda) = \hat{V}(x, \hat{s} \mid \mu, \mu, \lambda)$ for all $(x, \hat{s})$;
  2. $\lambda(\hat{s}) = \beta\, E_{\mu, \lambda}\big[\hat{V}(x^e, \hat{s}' \mid \mu, \mu, \lambda) \mid \hat{s}\big]$ for all $\hat{s}$.

- Same structure as the MPE, with s replaced by the much smaller $\hat{s}$.

- Because θ is not a sufficient statistic, MME strategies need not be best responses in the full game. Ifrach and Weintraub provide a computable **error bound** on the gain from a unilateral deviation to a strategy using the full state; if it is large, add more moments.

MME: Computational Algorithm

Algorithm 1 Equilibrium solver

1: Initialize with (μ, λ) and industry state $s_0 = (f_0, d_0, z_0)$ with corresponding $\hat{s}_0$;
2: $n := 1, \Delta := \epsilon + 1$;
3: **while** $\Delta > \epsilon$ **do**
4: Simulate a T period sample path $\{(f_t, d_t, z_t)\}_{t=1}^{T}$ with corresponding $\{\hat{s}_t\}_{t=1}^{T}$ for large T;
5: Calculate the observed frequencies of industry states $h(\hat{s}) := \frac{1}{T}\sum_{t=1}^{T} 1\{\hat{s}_t = \hat{s}\}$, for all $\hat{s} \in \hat{\mathcal{S}}$;
6: Use the same sample path to compute the perceived transition kernel $\hat{\mathbf{P}}_{\mu',\mu,\lambda}[\cdot|x, \hat{s}]$, for all $x, \hat{s}$;
7: Solve $\mu' := \underset{\tilde{\mu}\in\hat{\mathcal{M}}}{\text{argmax}}\, \hat{V}(x, \hat{s}|\tilde{\mu}, \mu, \lambda)$, for all $(x, \hat{s}) \in \mathcal{X} \times \hat{\mathcal{S}}$;
8: Let $\lambda'(\hat{s}) := \beta\, \mathrm{E}_{\mu,\lambda}[\hat{V}(x^e, \hat{s}_{t+1}|\mu', \mu, \lambda)|\hat{s}_t = \hat{s}]$, for all $\hat{s} \in \hat{\mathcal{S}}$;
9: $\Delta := \max(\|\mu - \mu'\|_h, \|\lambda - \lambda'\|_h)$;
10: $\mu := \mu + (\mu' - \mu)/(1 + n^{\sigma})$;
11: $\lambda := \lambda + (\lambda' - \lambda)/(1 + n^{\sigma})$;
12: $n := n + 1$;
13: **end while**;

- Lines 4–6: simulate a long path under the current strategies; use it to estimate the perceived kernel $\hat{P}$ (empirical frequencies of moment transitions).
- Lines 7–8: best responses and entry cutoffs given $\hat{P}$ — a small dynamic program on $\mathcal{X} \times \hat{\mathcal{S}}$.
- Lines 10–11: damped updates with step size $1/(1 + n^{\sigma})$; stop when strategies stop changing (norm weighted by visit frequencies h). Could be sped up further with Pakes–McGuire (2001) ideas.

Ifrach and Weintraub (2017): What We Learn

- In numerical experiments, MME closely approximates MPE where MPE can be computed, and remains tractable with many fringe firms where it cannot.
- **Fringe firms matter**: cumulatively, they discipline dominant firms to behave more competitively. Dominant firms may invest strategically to deter fringe growth and entry.
- A common shortcut in applied work — model only the few dominant firms and ignore the fringe (e.g., Ryan 2012) — can therefore give misleading counterfactuals.
- The same tools apply elsewhere: industry models with a continuum of firms and an aggregate shock, heterogeneous-agent macro models, and dynamic demand with forward-looking consumers.

How Severe Is Multiplicity? Besanko et al. (2010)

- Introduce **learning-by-doing** (LBD) and **organizational forgetting** (OF) into the EP framework, with two firms and pricing as the only control.

- LBD promotes market dominance: the more experienced firm can underprice its rival and shut out competition.
  - *Advantage-building*: winning a sale moves a firm down its own learning curve.
  - *Advantage-defending*: winning a sale also denies the rival the chance to move down its curve.
- One might think OF simply "undoes" LBD and is pro-competitive.

- **Surprising finding**: OF can make firms *more* aggressive.
  - Weak OF: LBD works as a ratchet; asymmetries arise but cannot persist.
  - Strong OF: forgetting stifles LBD altogether.
  - Intermediate OF: asymmetries arise and persist; the leader can delay or stall the follower more effectively. This is also where multiple equilibria appear.
- Forgetting matters because it allows **bidirectional** movement through the state space.

Besanko et al. (2010): States, Learning, and Forgetting

- Two firms $n \in \{1, 2\}$; know-how (experience) $e_n \in \{1, \ldots, M\}$ plays the role of ω. Industry state $\mathrm{e} = (e_1, e_2)$.
- Law of motion: $e_n' = e_n + q_n - f_n$, where $q_n \in \{0, 1\}$ indicates whether firm n makes the sale and $f_n \in \{0, 1\}$ whether it forgets a unit of know-how.

- **Learning-by-doing**: marginal cost falls along a learning curve until m,

$$c(e_n) = \begin{cases} \kappa\, e_n^{\eta}, & 1 \le e_n < m, \\ \kappa\, m^{\eta}, & m \le e_n \le M, \end{cases} \qquad \eta = \log_2 \rho.$$

  The **progress ratio** $\rho \in (0, 1]$: each doubling of experience multiplies cost by ρ (smaller ρ = steeper learning).

- **Organizational forgetting**: the probability of losing a unit of know-how is

$$\Delta(e_n) = 1 - (1 - \delta)^{e_n},$$

  increasing in e_n (e.g., the chance that at least one of e_n skilled workers leaves when each leaves with probability δ).
- Baseline: $M = 30$, $m = 15$, $\kappa = 10$, $\sigma = 1$, $\beta = 1/1.05$; (ρ, δ) varied over the whole range.

Besanko et al. (2010): Demand and Bellman Equation

- One buyer per period buys from one of the two firms (logit, no outside good):

$$D_n(\mathrm{p}) = \Pr(q_n = 1) = \frac{1}{1 + \exp\left(\frac{p_n - p_{-n}}{\sigma}\right)}.$$

- Transitions of know-how given the sale outcome:

$$\Pr(e_n' \mid e_n, q_n) = \begin{cases} 1 - \Delta(e_n), & e_n' = e_n + q_n, \\ \Delta(e_n), & e_n' = e_n + q_n - 1. \end{cases}$$

- **Bellman equation**:

$$V_n(\mathrm{e}) = \max_{p_n} \; D_n\big(p_n, p_{-n}(\mathrm{e})\big)\big(p_n - c(e_n)\big) + \beta \sum_{k=1}^{2} D_k\big(p_n, p_{-n}(\mathrm{e})\big)\, \bar{V}_{nk}(\mathrm{e}).$$

- $\bar{V}_{nk}(\mathrm{e})$ is firm n's expected value **given the buyer buys from firm** k:

$$\bar{V}_{n1}(\mathrm{e}) = \sum_{e_1'=e_1}^{e_1+1} \sum_{e_2'=e_2-1}^{e_2} V_n(\mathrm{e}')\Pr(e_1'|e_1,1)\Pr(e_2'|e_2,0),$$

$$\bar{V}_{n2}(\mathrm{e}) = \sum_{e_1'=e_1-1}^{e_1} \sum_{e_2'=e_2}^{e_2+1} V_n(\mathrm{e}')\Pr(e_1'|e_1,0)\Pr(e_2'|e_2,1).$$

- The price affects not only today's margin but also **who learns**: the term $D_1(\bar{V}_{n1} - \bar{V}_{n2})$ is the dynamic incentive to cut price.

Besanko et al. (2010): Equilibrium and Computation

- Firms are symmetric ex ante, so focus on **symmetric** MPE: $p_2(e_1, e_2) = p_1(e_2, e_1)$. It suffices to solve for $V \equiv V_1$ and $p \equiv p_1$.

- Bellman equation and first-order condition, for every state e:

$$F^1_{\mathrm{e}}(\mathrm{V}^*, \mathrm{p}^*) = -V^*(\mathrm{e}) + D^*_1(\mathrm{e})\big(p^*(\mathrm{e}) - c(e_1)\big) + \beta \sum_{k=1}^{2} D^*_k(\mathrm{e})\, \bar{V}^*_k(\mathrm{e}) = 0,$$

$$F^2_{\mathrm{e}}(\mathrm{V}^*, \mathrm{p}^*) = \sigma - \big(1 - D^*_1(\mathrm{e})\big)\big(p^*(\mathrm{e}) - c(e_1)\big) - \beta\, \bar{V}^*_1(\mathrm{e}) + \beta \sum_{k=1}^{2} D^*_k(\mathrm{e})\, \bar{V}^*_k(\mathrm{e}) = 0.$$

- The FOC uses the logit property $\partial D_1 / \partial p_1 = -D_1(1 - D_1)/\sigma$.
- This is a system of $2M^2 = 1{,}800$ nonlinear equations.

- Pakes–McGuire finds at most one equilibrium per starting point. Instead, Besanko et al. use a **homotopy**: fix ρ, start from the unique equilibrium at $\delta = 0$, and trace the solution path of $F(\mathrm{V}, \mathrm{p}; \delta) = 0$ as δ goes from 0 to 1. Where the path bends back, there are multiple equilibria.

Number of Equilibria

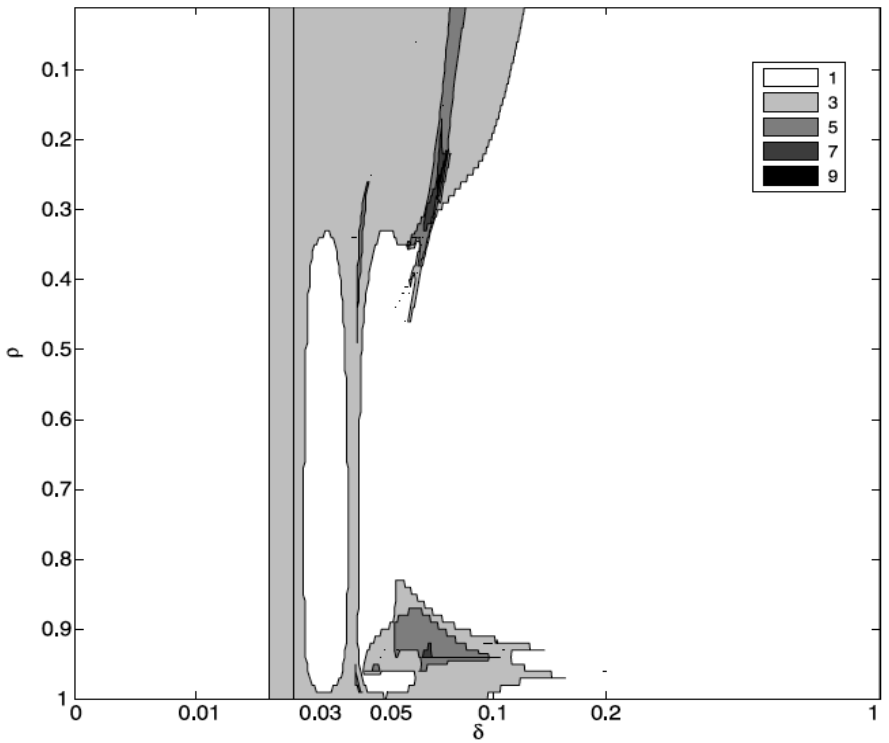


FIGURE 2.—Number of equilibria.

Number of symmetric equilibria found for each (forgetting rate δ, progress ratio ρ). Without forgetting ($\delta = 0$) the equilibrium is unique; multiplicity is concentrated at intermediate forgetting rates, including empirically plausible ones ($\rho \in [0.7, 0.95]$).

Policy Functions: Four Types of Equilibria

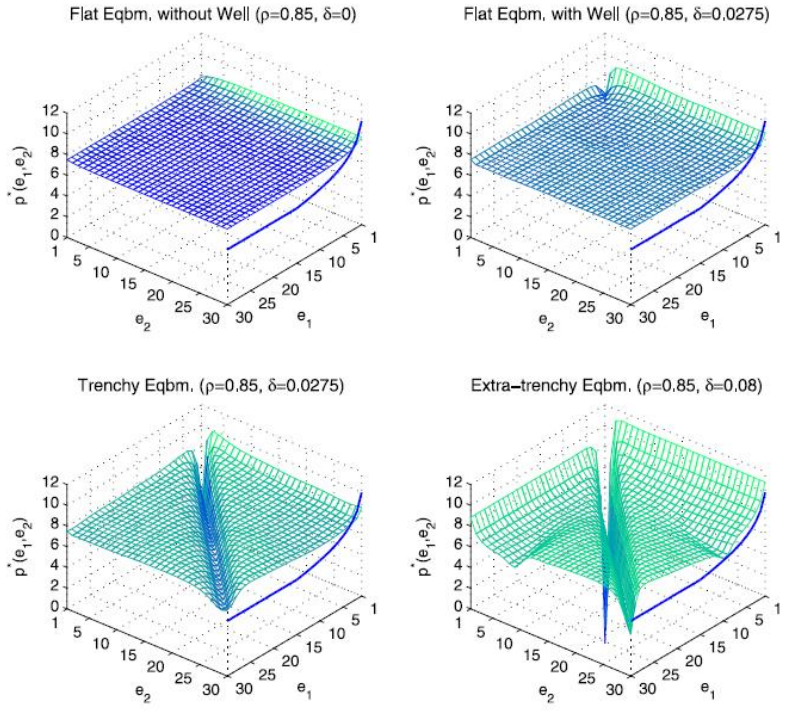
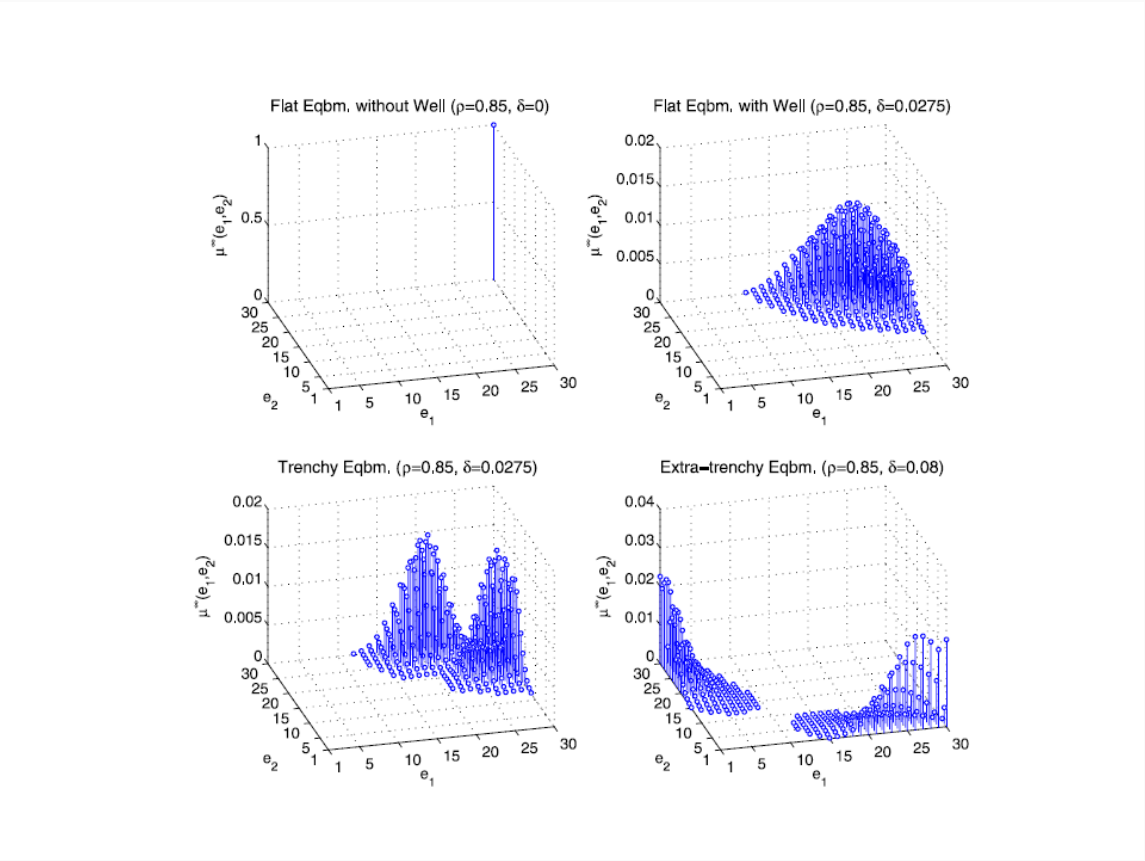


FIGURE 4.—Policy function $p^*(e_1, e_2)$; marginal cost $c(e_1)$ (solid line in $e_2 = 30$ plane).

Equilibrium price $p^*(e_1, e_2)$; the solid line is marginal cost. A **well**: deep price cuts near $(1, 1)$, a preemption battle to get ahead first. A **trench**: deep price cuts along the diagonal, a price war whenever the follower threatens to catch up. Note: the same $(\rho, \delta) = (0.85, 0.0275)$ supports both a flat and a trenchy equilibrium.

Long-Run Industry Structure



Limiting distribution $\mu^{\infty}(e_1, e_2)$ over industry states. Without forgetting both firms end at the bottom of their learning curves (a symmetric industry). In trenchy equilibria the mass sits off the diagonal: **persistent leadership**. With extra-trenchy pricing one firm is near-absolutely dominant.

Summary: Pricing Behaviour and Industry Dynamics

TABLE IV

PRICING BEHAVIOR AND INDUSTRY DYNAMICS

	Flat Eqbm. Without Well	Flat Eqbm. With Well	Trenchy Eqbm.	Extratrenchy Eqbm.
Leading example	$\rho = 0.85$, $\delta = 0$	$\rho = 0.85$, $\delta = 0.0275$	$\rho = 0.85$, $\delta = 0.0275$	$\rho = 0.85$, $\delta = 0.08$
Preemption battle (well)	no	yes	no	no
Price war triggered by imminent threat (diagonal trench)	no	no	yes	yes
Price war triggered by distant threat (sideways trench)	no	no	no	yes
Short-run market dominance	no	yes	yes	yes
Long-run market dominance	no	no	yes, modest	yes, extreme
Dominance properties	yes	no, mostly	no, mostly	no, mostly

- The **equilibrium itself**, not just primitives, determines whether the industry ends up symmetric or dominated.
- For estimation and counterfactuals: we need to know which equilibrium is played in the data and whether a policy change moves the industry to another one.

Summary

- **Facts**: high infant mortality, exit falling with age and size, fast growth of young survivors, persistent productivity dispersion, a heavy-tailed size distribution.
- **Selection**: Jovanovic (learning about a fixed type) and Hopenhayn (persistent productivity shocks) explain these facts as outcomes of market selection. Hopenhayn's stationary equilibrium is cheap to compute because the aggregate state is deterministic.
- **Sunk costs** shape turnover, selection, and aggregate productivity (Aw–Chung–Roberts).
- **Strategic dynamics**: Ericson–Pakes turns these ideas into a computable dynamic game. The price is the curse of dimensionality and, often, multiple equilibria.
- **Computation**: Pakes–McGuire (1994) is transparent but scales badly; stochastic algorithms, continuous time, and alternative equilibrium concepts (OE, MME) extend the reach.

Next lecture: estimation

How do we take these models to data without solving for equilibrium at every parameter value? Two-step (CCP) estimators: Aguirregabiria and Mira (2007); Bajari, Benkard, and Levin (2007); Pakes, Ostrovsky, and Berry (2007); applications to entry and exit (Collard-Wexler 2013; Dunne, Klimek, Roberts, and Xu 2013).

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